



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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Class : 10

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COMMON HALF YEARLY EXAMINATION - 2024 - 25

Time Allowed : 3.00 Hours]

SCIENCE

[Max. Marks : 75

PART - I

I. Note: (i) Answer all the questions.

(ii) Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer. 12x1=12

1. The unit of 'g' is ms^{-2} . It can be also expressed as
a) Cms^{-1} b) Nkg^{-1} c) $\text{Nm}^2\text{kg}^{-1}$ d) Cm^2s^{-2}
2. Power of a lens is -4D, then its focal length is
a) 4m b) -40m c) -0.25m d) -2.5 m
3. ----- aprons are used to Protect us from gamma radiations.
a) Lead oxide b) Iron c) Lead d) Aluminium
4. The Volume occupied by 1 mole of diatomic gas at S.T.P. is
a) 11.2 litre b) 5.6 litre c) 22.4 litre d) 44.8 litre
5. Which of the following is the universal solvent?
a) Acetone b) Benzene c) Water d) Alcohol
6. TFM in soaps represents ----- content in soap.
a) Mineral b) Vitamin c) Fatty acid d) Carbohydrate
7. Who discovered chemical pathway for photo synthesis
a) C.N.R. Rao b) Kolliker c) Sochs d) Melvin Calvin
8. The brain of leech lies above the
a) mouth b) Buccal cavity c) Pharynx d) Crop
9. Nerve cells do not possess
a) neurilemma b) sarcolemma c) axon d) dendrites
10. Which one of the following is an IUCD?
a) Copper T b) Oral pills c) Diaphragm d) Tubectomy
11. The centromere is found at the centre of the ----- chromosome.
a) Telecentric b) Metacentric c) Sub-metacentric d) Acrocentric
12. All files are stored in the -----
a) Folder b) box c) paint d) Scanner

Part - II

II. Note: (i) Answer any seven questions.

(ii) Q.No. 22 is compulsory.

7x2=14

13. State Newton's second law.
14. Draw a ray diagram to show the image formed by a Convex lens when the object is placed between F and 2F.
15. State Joule's law of heating.
16. Define atomicity.
17. Fill in the blanks.
a) The general molecular formula of Alkyne is -----
b) 100% pure Alcohol is called -----



18. Who discovered Rh factor? Why was it named so?
19. What are Synthetic auxins? Give example?
20. Name the secondary sex organs in male.
21. What is metastasis?
22. A radon specimen emits radiations of 3.7×10^3 G Bq per second. Convert this disintegration in term of Curie.

PART - III

III. Note: Answer any seven questions. Q.No: 32 is compulsory.

7x4 = 28

23. List any four Properties of light.
24. a) Define the unit of Current.
b) Distinguish between resistivity and conductivity of a conductor.
25. a) What is meant by alloys
b) What is meant by binary solution
26. In What way hygroscopic substances differ from deliquescent substances.
27. Difference between monocot root and Dicot root
28. How do plants absorb water? Explain.
29. Name the gaseous plant hormone. Describe its three different actions in plants.
30. Define Ethanobotany and write its importance.
31. Enumerate the importance of forest.
32. a) When an aqueous solution of Potassium Chloride is added to an aqueous solution of Silver Nitrate, a white Precipitate is formed. Give the chemical equation of this reaction.
b) If the P^H of a solution is 4.5, what is its P^{OH} ?

PART - IV

IV. Note: Answer all the questions.

3x7=21

Draw diagram wherever necessary.

33. a) i) Define one Calorie.
ii) State and Prove the law of conservation of linear momentum.
(OR)
b) i) Mention two cases in which there is no Doppler effect in sound?
ii) Explain the process of controlled and uncontrolled chain reactions.
34. a) (i) A is a Silvery White Metal. A Combines with O_2 to form B at $800^\circ C$, the alloy of A is used in making the aircraft. Find A and B.
(ii) Give the Salient features of "Modern Atomic Theory".
(OR)
b) i) Differentiate reversible and irreversible reactions.
ii) How is Ethanol manufactured from Sugar Cane?
35. a) i) Write the dental formula of rabbit.
ii) Classify neurons based on its structure with suitable diagram.
(OR)
b) i) State the applications of DNA finger Printing technique.
ii) How is the structure of DNA Organised? What is the biological Significance of DNA?